Band-1 Hadamard Order 3316 — Miyamoto (H_828 Paley-I) + Paley-II; Gap 2052 Still OPEN

H_828/C_830/H_3316 PASS; 3316 in SPK via Paley-II; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_3284_Miyamoto`; `2026-09-15_Band1_Hadamard_3188_Miyamoto`; `2026-09-15_Band1_Hadamard_3076_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_3316_miyamoto/`.
Cutoff. 2026-09-15T10:49:40Z (~6:49AM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 828 and 3316 only** (verified max $|HH^\top - nI| = 0$), via Miyamoto with Paley type I seed, and independently via Paley type II. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 3316 **enters SPK via Paley-II** ($q = 1657$ prime $\equiv 1 \pmod 4$; order $2(q+1) = 3316$): Paley-I **FAIL** ($q = 3315$ not prime); Kronecker **FAIL** ($3316 = 4 \cdot 829$, 829 not Hadamard-admissible). Seed H_{828} uses Paley-I ($q = 827$); Paley-II seed for 828 **FAIL** ($2(p+1) = 828 \Rightarrow p = 413$ not prime). `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 3284 on box. This increment ships the next Miyamoto target: **order 3316** ($3316 = 4 \cdot 829$, 829 prime $\equiv 1 \pmod 4$). Seed H_{828} via Paley type I ($q = 827$ prime $\equiv 3 \pmod 4$; order $q+1 = 828$); Paley-II seed path for 828 **rejected** ($p = 413$ not prime). Symmetric Paley conference C_{830} over prime 829. $H_{3316} = \text{Miyamoto}(H_{828}, C_{830})$. Independently, H_{3316} via Paley type II ($q = 1657$). Unit tests **25/25 PASS**. SPK next gap remains **2052** (OPEN). Order 3316 **is in SPK** via Paley-II (not Miyamoto-only). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 3316; Miyamoto; Paley type I; Paley type II; Paley conference; constructive gap 2052; SPK; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{828} constructed and verified (Paley type I, $q = 827$). 2. C_{830} constructed and verified (Paley conference, $q = 829$): $CC^\top = 829I$, zero diagonal. 3. H_{3316} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 4. H_{3316} also constructed and verified (Paley type II, $q = 1657$) — SPK classical path. 5. SPK: $3316 \in$ SPK via Paley-II. Paley-I/Kronecker paths for 3316 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$3316 = 4 \cdot 829$
$q = 829$	prime, $829 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{828}
Needed conference	order $830 = q + 1$
H_{828} path (used)	Paley type I, $q = 827$ (prime $\equiv 3 \pmod 4$); $q + 1 = 828$
H_{828} Paley-II	**REJECTED**: $2(p + 1) = 828 \Rightarrow p = 413$ not prime
Conference path	Symmetric Paley over prime 829
Paley-I for 3316	**REJECTED**: $q' = 3315$ not prime
Paley-II for 3316	**PASS**: $q' = 1657$ prime $\equiv 1 \pmod 4$; $2(q' + 1) = 3316$
Kronecker for 3316	**REJECTED**: only 4×829; $829 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{828}	Paley type I, $q = 827$	0	0.6127	**PASS**
C_{830}	Paley conference, $q = 829$	0	0.5791	**PASS**
H_{3316}	Miyamoto(H_{828}, C_{830})	0	15.3095	**PASS**
H_{3316}	Paley type II, $q = 1657$	0	16.8560	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh + Paley-II) ≈ 34.9236 s. Compressed ± 1 matrices saved as `H828_pm1.npz`, `C830_pm1.npz`, `H3316_pm1.npz`, `H3316_paleyII_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible $\leq 2000 \cup$ on-box prior ships (through 3284/820/...) $\cup$ verified $H_{828} \cup$ Paley-II order 3316 $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 3316 enters SPK by the classical Paley-II path.

Quantity	Value
Known count	886
3316 in SPK	**True** (via Paley-II)
828 in SPK	**True** (via Paley-I)
Next constructive gap	**2052** (OPEN)
Miyamoto-only bonus?	**No** (unlike 3284/3076)

5. Gap 2052 (OPEN; method triage / non-finding ONLY)

Path	Result
Paley-I ($q = 2051$)	FAIL
Paley-II ($q = 1025$)	FAIL
Kronecker	FAIL ($2052 = 2^2 \cdot 3^3 \cdot 19$)
Miyamoto + Paley conference ($q = 513$)	FAIL (513 not prime)
GS / SDS / Williamson / Yamada	lit-only; not constructed this run

Status: OPEN. Non-finding $\neq$ counterexample. Conjecture remains OPEN.

6. Unit tests

25/25 PASS (Sylvester/Paley/Miyamoto controls; $q = 827/829/1657$ arithmetic; SPK path eligibility; reject-all-ones; conference control; gap-2052-not-filled guard).

7. Non-claims

- Full Hadamard conjecture
- Clay-closed / finished-ToE
- Gap 2052 filled
- RH / P≠NP / YM / Frankl / NS-solved / AC-solved
- Invented Object 1 / J
- GATE-04 unlocked

8. Next optional

- Method-attack SPK gap **2052** (GS/SDS / Yamada / Williamson)
- Next Miyamoto: $q = 853 \to n = 3412$ (seed notes sparse) or $q = 857 \to n = 3428$ (Kronecker 428×2) or $q = 881 \to n = 3524$
- Track B privacy / Object 1 meter await (GATE-04 LOCKED)

9. Artifacts

- Lab: `/workspace/experiments/band1/hadamard_3316_miyamoto/`
- `verify_hadamard_3316.py`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`
- Matrices: `H828_pm1.npz`, `C830_pm1.npz`, `H3316_pm1.npz`, `H3316_paleyII_pm1.npz`
- This note: `2026-09-15_Band1_Hadamard_3316_Miyamoto.md` / `.pdf`

Claim level. Finite constructions of orders **828** and **3316** ONLY. **3316 in SPK via Paley-II.** Gap **2052 OPEN**. Conjecture **OPEN**. GATE-04 **LOCKED**. `clay_closed=false`.